

**PACKAGE INSERT – INSTRUCTIONS FOR USE – READ
CAREFULLY!**

**DEXMEDETOMIDINE FRESENIUS 100 MCG/ML
CONCENTRATE FOR SOLUTION FOR INFUSION**

PRODUCT NAME

Dexmedetomidine Fresenius 100 mcg/ml Concentrate for Solution for Infusion

NAME AND STRENGTH OF ACTIVE INGREDIENT (S)

Dexmedetomidine as Dexmedetomidine Hydrochloride and 100mcg/ml

Each 2 ml vial contains 200 micrograms of dexmedetomidine.

Refer to Section 14 for List of Excipients.

PRODUCT DESCRIPTION

Dexmedetomidine Fresenius 100 mcg/ml is a sterile, clear, colourless to slightly brownish concentrate for solution for infusion filled in a clear glass vial. Sodium chloride is used as tonicity adjusting agent and water for injection as solvent.

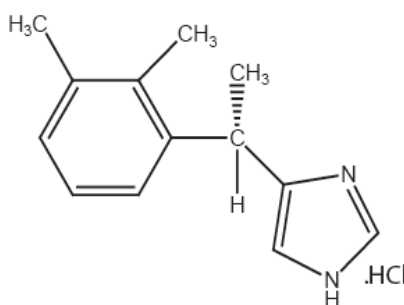
Dexmedetomidine 100 mcg/ml is intended for I.V infusion use after dilution with diluent, 0.9% sodium chloride.

Diluted Dexmedetomidine 100 mcg/ml is a clear, colourless solution.

CHEMICAL STRUCTURE

Dexmedetomidine Hydrochloride is chemically designated as 4-[(1S)-1-(2, 3-Dimethylphenyl)ethyl]-1H-imidazole hydrochloride. Dexmedetomidine Hydrochloride is a white crystalline powder. It is freely soluble in water and alcohols and slightly soluble in methylene chloride and acetone.

Its molecular formula is $C_{13}H_{16}N_2 \cdot HCl$ and its molecular weight is 236.7. The structural formula is as follows:



PHARMACOLOGY

Mechanism of Action

Dexmedetomidine is a relatively selective centrally acting α_2 -adrenergic agonist with sedative properties. α_2 selectivity is observed in animals following slow intravenous infusion of low and medium doses (10-300 mcg/kg). Both α_1 and α_2 activity is observed following slow intravenous infusion of high doses (≥ 1000 mcg/kg) or with rapid intravenous administration.

Pharmacodynamic

In a study in healthy volunteers (N=10), respiratory rate and oxygen saturation remained within

normal limits and there was no evidence of respiratory depression when Dexmedetomidine was administered by intravenous infusion at doses within the recommended dose range (0.2 - 0.7 mcg/kg/hr).

Pharmacokinetic properties

Following intravenous administration, dexmedetomidine exhibits the following pharmacokinetic parameters: a rapid distribution phase with a distribution half-life ($t_{1/2}$) of approximately 6 minutes; a terminal elimination half-life ($t_{1/2}$) of approximately 2 hours; and steady-state volume of distribution (V_{ss}) of approximately 118 liters. Clearance is estimated to be approximately 39 L/h. The mean body weight associated with this clearance estimate was 72 kg.

Dexmedetomidine exhibits linear pharmacokinetics in the dosage range of 0.2 to 0.7 mcg/kg/hr when administered by intravenous infusion for up to 24 hours. Table 8 shows the main pharmacokinetic parameters when dexmedetomidine was infused (after appropriate loading doses) at maintenance infusion rates of 0.17 mcg/kg/hr (target plasma concentration of 0.3 ng/mL) for 12 and 24 hours, 0.33 mcg/kg/hr (target plasma concentration of 0.6 ng/mL) for 24 hours, and 0.70 mcg/kg/hr (target plasma concentration of 1.25 ng/mL) for 24 hours.

Table 8. Mean $\pm$ SD Pharmacokinetic Parameters				
Parameter	Loading Infusion (min)/Total infusion duration (hrs)			
	10 min/12 hrs	10 min/24 hrs	10 min/24 hrs	35 min/24 hrs
	Dexmedetomidine Target Plasma Concentration (ng/mL) and Dose (mcg/kg/hr)			
	0.3/0.17	0.3/0.17	0.6/0.33	1.25/0.70
$t_{1/2}$ *, hour	1.78 $\pm$ 0.30	2.22 $\pm$ 0.59	2.23 $\pm$ 0.21	2.50 $\pm$ 0.61
CL, liter/hour	46.3 $\pm$ 8.3	43.1 $\pm$ 6.5	35.3 $\pm$ 6.8	36.5 $\pm$ 7.5
V_{ss} , liter	88.7 $\pm$ 22.9	102.4 $\pm$ 20.3	93.6 $\pm$ 17.0	99.6 $\pm$ 17.8
Avg C_{ss} #, ng/mL	0.27 $\pm$ 0.05	0.27 $\pm$ 0.05	0.67 $\pm$ 0.10	1.37 $\pm$ 0.20

* Presented as harmonic mean and pseudo standard deviation.

Mean C_{ss} = Average steady-state concentration of dexmedetomidine. The mean C_{ss} was calculated based on post-dose sampling from 2.5 - 9 hours samples for 12 hour infusion and post-dose sampling from 2.5 - 18 hours for 24 hour infusions.

The loading doses for each of the above indicated groups were 0.5, 0.5, 1 and 2.2 mcg/kg, respectively.

Dexmedetomidine pharmacokinetic parameters after Dexmedetomidine maintenance doses of 0.2 to 1.4 mcg/kg/hr for >24 hours were similar to the pharmacokinetic (PK) parameters after Dexmedetomidine maintenance dosing for <24 hours in other studies. The values for clearance (CL), volume of distribution (V), and $t_{1/2}$ were 39.4 L/hr, 152 L, and 2.67 hours, respectively.

Distribution

The steady-state volume of distribution (V_{ss}) of dexmedetomidine was approximately 118 liters. Dexmedetomidine protein binding was assessed in the plasma of normal healthy male and female subjects. The average protein binding was 94% and was constant across the different plasma concentrations tested. Protein binding was similar in males and females. The fraction of Dexmedetomidine that was bound to plasma proteins was significantly decreased in subjects with hepatic impairment compared to healthy subjects.

The potential for protein binding displacement of dexmedetomidine by fentanyl, ketorolac, theophylline, digoxin and lidocaine was explored in vitro, and negligible changes in the plasma protein binding of Dexmedetomidine were observed. The potential for protein binding displacement of phenytoin, warfarin, ibuprofen, propranolol, theophylline and digoxin by Dexmedetomidine was explored in vitro and none of these compounds appeared to be significantly displaced by Dexmedetomidine.

Elimination

Metabolism

Dexmedetomidine undergoes almost complete biotransformation with very little unchanged dexmedetomidine excreted in urine and feces. Biotransformation involves both direct glucuronidation as well as cytochrome P450 mediated metabolism. The major metabolic pathways of dexmedetomidine are: direct N-glucuronidation to inactive metabolites; aliphatic hydroxylation (mediated primarily by CYP2A6 with a minor role of CYP1A2, CYP2E1, CYP2D6 and CYP2C19) of dexmedetomidine to generate 3-hydroxydexmedetomidine, the glucuronide of 3-hydroxy-dexmedetomidine, and 3-carboxy dexmedetomidine; and N-methylation of dexmedetomidine to generate 3-hydroxy N-methyldexmedetomidine, 3-carboxy N-methyl-dexmedetomidine, and dexmedetomidine-N-methyl Oglucuronide.

Excretion

The terminal elimination half-life ($t_{1/2}$) of dexmedetomidine is approximately 2 hours and clearance is estimated to be approximately 39 L/h. A mass balance study demonstrated that after nine days an average of 95% of the radioactivity, following intravenous administration of radio labelled dexmedetomidine, was recovered in the urine and 4% in the feces. No unchanged dexmedetomidine was detected in the urine. Approximately 85% of the radioactivity recovered in the urine was excreted within 24 hours after the infusion. Fractionation of the radioactivity excreted in urine demonstrated that products of N-glucuronidation accounted for approximately 34% of the cumulative urinary excretion. In addition, aliphatic hydroxylation of parent drug to form 3-hydroxy-dexmedetomidine, the glucuronide of 3-hydroxydexmedetomidine, and 3-carboxylic acid-dexmedetomidine together represented approximately 14% of the dose in urine. N-methylation of dexmedetomidine to form 3 hydroxy N-methyl dexmedetomidine, 3-carboxy N-methyl dexmedetomidine, and N-methyl O-glucuronide dexmedetomidine accounted for approximately 18% of the dose in urine. The N-Methyl metabolite itself was a minor circulating component and was undetected in urine. Approximately 28% of the urinary metabolites have not been identified.

Specific Populations

Gender

There was no observed difference in Dexmedetomidine pharmacokinetics due to gender.

Geriatrics

The pharmacokinetic profile of Dexmedetomidine was not altered by age. There were no differences in the pharmacokinetics of Dexmedetomidine in young (18 - 40 years), middle age (41 - 65 years), and elderly (>65 years) subjects.

Hepatic Impairment

In subjects with varying degrees of hepatic impairment (Child-Pugh Class A, B, or C), clearance values for Dexmedetomidine were lower than in healthy subjects. The mean clearance values for patients with mild, moderate, and severe hepatic impairment were 74%, 64% and 53% of those observed in the normal healthy subjects, respectively. Mean clearances for free drug were 59%, 51% and 32% of those observed in the normal healthy subjects, respectively.

Although Dexmedetomidine is dosed to effect, it may be necessary to consider dose reduction in subjects with hepatic impairment (see *Dosage and Administration, Warnings and Precautions*).

Renal Impairment

Dexmedetomidine pharmacokinetics (C_{max} , T_{max} , AUC, $t_{1/2}$, CL, and V_{ss}) were not significantly different in patients with severe renal impairment (creatinine clearance: <30 mL/min) compared to healthy subjects.

Drug Interactions

In vitro studies: *In vitro* studies in human liver microsomes demonstrated no evidence of cytochrome P450 mediated drug interactions that are likely to be of clinical relevance.

INDICATIONS AND USAGE

Intensive Care Unit Sedation:

Dexmedetomidine Fresenius is indicated for sedation of initially intubated and mechanically ventilated patients during treatment in an intensive care setting. Dexmedetomidine Fresenius should be administered by continuous infusion not to exceed 24 hours.

Dexmedetomidine Fresenius has been continuously infused in mechanically ventilated patients prior to extubation, during extubation, and post-extubation. It is not necessary to discontinue Dexmedetomidine Fresenius prior to extubation.

Procedural Sedation:

Dexmedetomidine Fresenius is indicated for sedation of non-intubated patients prior to and/or during surgical and other procedures.

ROUTE OF ADMINISTRATION

Intravenous Infusion use.

Dexmedetomidine Fresenius must be administered only as a diluted intravenous infusion using a controlled infusion device.

DOSAGE AND ADMINISTRATION

Dosing Guidelines

- Dexmedetomidine dosing should be individualized and titrated to desired clinical response.
- Dexmedetomidine is not indicated for infusions lasting longer than 24 hours
- Dexmedetomidine should be administered using a controlled infusion device.

Table 1. Dosage Information

INDICATION	DOSAGE AND ADMINISTRATION
Initiation of Intensive Care Unit Sedation:	For adult patients: a loading infusion of one mcg/kg over 10 minutes. For patients over 65 years of age: a dose reduction should be considered (see <i>Use in Special Populations</i>) For patients with impaired hepatic or renal function: a dose reduction should be considered (see <i>Use in Special Populations, Pharmacology</i>).
Maintenance of Intensive Care Unit Sedation:	For adult patients: a maintenance infusion of 0.2 to 0.7 mcg/kg/hr. The rate of the maintenance infusion should be adjusted to achieve the desired level of sedation.

	For patients over 65 years of age: a dose reduction should be considered (<i>see Use in Special Populations</i>) For patients with impaired hepatic or renal function: a dose reduction should be considered (<i>Use in Special Populations, Pharmacology</i>).
Initiation of Procedural Sedation:	For adult patients: a loading infusion of one mcg/kg over 10 minutes. For less invasive procedures such as ophthalmic surgery, a loading infusion of 0.5 mcg/kg given over 10 minutes may be suitable. For awake fiberoptic intubation patients: a loading infusion of one mcg/kg over 10 minutes. For patients over 65 years of age: a loading infusion of 0.5 mcg/kg over 10 minutes (<i>see Use in Special Populations</i>). For patients with impaired hepatic or renal function: a dose reduction should be considered (<i>see Use in Special Populations, Pharmacology</i>).
Maintenance of Procedural Sedation:	For adult patients: the maintenance infusion is generally initiated at 0.6 mcg/kg/hr and titrated to achieve desired clinical effect with doses ranging from 0.2 to 1 mcg/kg/hr. The rate of the maintenance infusion should be adjusted to achieve the targeted level of sedation. For awake fiberoptic intubation patients: a maintenance infusion of 0.7 mcg/kg/hr is recommended until the endotracheal tube is secured. For patients over 65 years of age: a dose reduction should be considered (<i>see Use in Special Populations</i>). For patients with impaired hepatic or renal function: a dose reduction should be considered (<i>see Use in Special Populations, Pharmacology</i>).

Dosage Adjustment

Due to possible pharmacodynamic interactions, a reduction in dosage of Dexmedetomidine or other concomitant anesthetics, sedatives, hypnotics or opioids may be required when co-administered. (*see Drug Interactions*).

Dosage reductions may need to be considered for patients with renal or hepatic impairment, and geriatric patients (*see Warnings and Precautions, Use in Special Populations, Pharmacology*).

Preparation of Solution

Dexmedetomidine Fresenius must be diluted in 0.9% sodium chloride solution to achieve required concentration (4 mcg/mL) prior to administration. Preparation of solutions is the same, whether for the loading dose or maintenance infusion.

Strict aseptic technique must always be maintained during handling of Dexmedetomidine Fresenius.

To prepare the infusion, withdraw 2 mL of Dexmedetomidine Fresenius and add to 48 mL of 0.9% sodium chloride injection to a total of 50 mL. Shake gently to mix well.

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit. Do not use if product is discolored or if precipitate matter is present.

Administration with Other Fluids

Dexmedetomidine Fresenius infusion should not be co-administered through the same intravenous catheter with blood or plasma because physical compatibility has not been established.

Dexmedetomidine Fresenius has been shown to be incompatible when administered with the following drugs: amphotericin B, diazepam.

Dexmedetomidine Fresenius has been shown to be compatible when administered with the following intravenous fluids: 0.9% sodium chloride in water, 5% dextrose in water, 20% mannitol, Lactated Ringer's solution, 100 mg/mL magnesium sulfate solution and 0.3% potassium chloride solution.

Compatibility

Compatibility with Natural Rubber

Compatibility studies have shown potential for adsorption of dexmedetomidine to some types of natural rubber. Although dexmedetomidine is dosed to effect, it is advisable to use components with synthetic or coated natural rubber gaskets.

Incompatibilities

Dexmedetomidine infusion should not be co-administered through the same intravenous catheter with blood, serum, or plasma because physical compatibility has not been established. Dexmedetomidine has been shown to be incompatible when administered with the following drugs: amphotericin B, diazepam.

CONTRAINDICATIONS

Patients who are hypersensitive to this drug or to any ingredient in the formulation component of the container.

WARNINGS AND PRECAUTIONS

Drug Administration

Dexmedetomidine should be administered only by persons skilled in the management of patients in the intensive care or operating room setting. Due to the known pharmacological effects of Dexmedetomidine, patients should be continuously monitored while receiving Dexmedetomidine.

Hypotension, Bradycardia, and Sinus Arrest

Clinically significant episodes of bradycardia and sinus arrest have been reported with Dexmedetomidine administration in young, healthy adult volunteers with high vagal tone or with different routes of administration including rapid intravenous or bolus administration.

Reports of hypotension and bradycardia have been associated with Dexmedetomidine infusion. Some of these cases have resulted in fatalities. If medical intervention is required, treatment may include decreasing or stopping the infusion of Dexmedetomidine, increasing the rate of intravenous fluid administration, elevation of the lower extremities, and use of pressor agents. Because Dexmedetomidine has the potential to augment bradycardia induced by vagal stimuli, clinicians should be prepared to intervene. The intravenous administration of anticholinergic agents (e.g., glycopyrrolate, atropine) should be considered to modify vagal tone. In clinical trials, glycopyrrolate or atropine were effective in the treatment of most episodes of Dexmedetomidine-induced bradycardia. However, in some patients with significant cardiovascular dysfunction, more advanced resuscitative measures were required.

Caution should be exercised when administering Dexmedetomidine to patients with advanced heart block and/or severe ventricular dysfunction. Because Dexmedetomidine decreases

sympathetic nervous system activity, hypotension and/or bradycardia may be expected to be more pronounced in patients with hypovolemia, diabetes mellitus, or chronic hypertension and in elderly patients.

In clinical trials where other vasodilators or negative chronotropic agents were co-administered with Dexmedetomidine an additive pharmacodynamic effect was not observed. Nonetheless, caution should be used when such agents are administered concomitantly with Dexmedetomidine.

Transient Hypertension

Transient hypertension has been observed primarily during the loading dose in association with the initial peripheral vasoconstrictive effects of Dexmedetomidine. Treatment of the transient hypertension has generally not been necessary, although reduction of the loading infusion rate may be desirable.

Arousability

Some patients receiving Dexmedetomidine have been observed to be arousable and alert when stimulated. This alone should not be considered as evidence of lack of efficacy in the absence of other clinical signs and symptoms.

Withdrawal

Intensive Care Unit Sedation

With administration up to 7 days, regardless of dose, 12 (5%) Dexmedetomidine adult subjects experienced at least 1 event related to withdrawal within the first 24 hours after discontinuing study drug and 7 (3%) Dexmedetomidine adult subjects experienced at least 1 event 24 to 48 hours after end of study drug. The most common events were nausea, vomiting, and agitation. In adult subjects, tachycardia and hypertension requiring intervention in the 48 hours following study drug discontinuation occurred at frequencies of <5%. If tachycardia and/or hypertension occurs after discontinuation of Dexmedetomidine supportive therapy is indicated.

Procedural Sedation

In adult subjects, withdrawal symptoms were not seen after discontinuation of short-term infusions of Dexmedetomidine (<6 hours).

Tolerance and Tachyphylaxis

Use of dexmedetomidine beyond 24 hours has been associated with tolerance and tachyphylaxis and a dose-related increase in adverse reactions (*see Adverse Reactions*).

Hyperthermia or Pyrexia

Dexmedetomidine may induce hyperthermia or pyrexia, which may be resistant to traditional cooling methods, such as administration of cooled intravenous fluids and antipyretic medications. Discontinue Dexmedetomidine if drug-related hyperthermia or pyrexia is suspected and monitor patients until body temperature normalizes.

Hepatic Impairment

Since Dexmedetomidine clearance decreases with severity of hepatic impairment, dose reduction should be considered in patients with impaired hepatic function (*see Dosage and Administration*).

Use In Specific Populations

Pediatric Use

Safety and efficacy have not been established for Procedural or ICU Sedation in pediatric patients. One assessor-blinded trial in pediatric patients and two open label studies in neonates were conducted to assess efficacy for ICU sedation. These studies did not meet their primary efficacy endpoints and the safety data submitted were insufficient to fully characterize the safety profile of Dexmedetomidine for this patient population. The use of Dexmedetomidine for procedural sedation in pediatric patients has not been evaluated.

Geriatric Use

Intensive Care Unit Sedation

A total of 729 patients in the clinical studies were 65 years of age and over. A total of 200 patients were 75 years of age and over. In patients greater than 65 years of age, a higher incidence of bradycardia and hypotension was observed following administration of Dexmedetomidine (*see Warnings and Precautions*). Therefore, a dose reduction may be considered in patients over 65 years of age (*see Dosage and Administration (2.2) and Clinical Pharmacology*).

Procedural Sedation

A total of 131 patients in the clinical studies were 65 years of age and over. A total of 47 patients were 75 years of age and over. Hypotension occurred in a higher incidence in Dexmedetomidine treated patients 65 years or older (72%) and 75 years or older (74%) as compared to patients <65 years (47%). A reduced loading dose of 0.5 mcg/kg given over 10 minutes is recommended and a reduction in the maintenance infusion should be considered for patients greater than 65 years of age.

Hepatic Impairment

Since Dexmedetomidine clearance decreases with increasing severity of hepatic impairment, dose reduction should be considered in patients with impaired hepatic function (*see Dosage and Administration and Clinical Pharmacology*).

Drug Abuse and Dependence

Controlled Substance

Dexmedetomidine (dexmedetomidine hydrochloride) is not a controlled substance.

Dependence

The dependence potential of Dexmedetomidine has not been studied in humans. However, since studies in rodents and primates have demonstrated that Dexmedetomidine exhibits pharmacologic actions similar to those of clonidine, it is possible that Dexmedetomidine may produce a clonidine-like withdrawal syndrome upon abrupt discontinuation (*see Warnings and Precautions*).

INTERACTIONS WITH OTHER MEDICAMENTS

Anesthetics, Sedatives, Hypnotics, Opioids

Co-administration of Dexmedetomidine with anesthetics, sedatives, hypnotics, and opioids is likely to lead to an enhancement of effects. Specific studies have confirmed these effects with sevoflurane, isoflurane, propofol, alfentanil, and midazolam. No pharmacokinetic interactions between Dexmedetomidine and isoflurane, propofol, alfentanil and midazolam have been demonstrated. However, due to possible pharmacodynamic interactions, when co-administered with Dexmedetomidine, a reduction in dosage of Dexmedetomidine or the concomitant anesthetic, sedative, hypnotic or opioid may be required.

Neuromuscular Blockers

In one study of 10 healthy adult volunteers, administration of Dexmedetomidine for 45 minutes at a plasma concentration of one ng/mL resulted in no clinically meaningful increases in the magnitude of neuromuscular blockade associated with rocuronium administration.

FERTILITY, PREGNANCY AND LACTATION

Pregnancy

Risk Summary

Available data from published randomized controlled trials and case reports over several decades of use with intravenously administered dexmedetomidine during pregnancy have not identified a drug associated risk of major birth defects and miscarriage; however, the reported exposures occurred after the first trimester. Most of the available data are based on studies with exposures that occurred at the time of caesarean section delivery, and these studies have not identified an adverse effect on maternal outcomes or infant Apgar scores. Available data indicate that dexmedetomidine crosses the placenta.

In animal reproduction studies, fetal toxicity that lower fetal viability and reduced live fetuses occurred with subcutaneous administration of dexmedetomidine to pregnant rats during organogenesis at doses 1.8 times the maximum recommended human dose (MRHD) of 17.8 mcg/kg/day.

Developmental toxicity (low pup weights and adult offspring weights, decreased F1 grip strength, increased early implantation loss and decreased viability of second-generation offspring) occurred when pregnant rats were subcutaneously administered dexmedetomidine at doses less than the clinical dose from late pregnancy through lactation and weaning (*see Data*).

The estimated background risk of major birth defects and miscarriage for the indicated population is unknown. All pregnancies have a background risk of birth defect, loss, or other adverse outcomes. In the U.S. general population, the estimated background risk of major birth defects and miscarriage in clinically recognized pregnancies is 2-4% and 15-20%, respectively.

Data

Animal Data

Increased post-implantation losses and reduced live fetuses in the presence of maternal toxicity (i.e. decreased body weight) were noted in a rat embryo-fetal development study in which pregnant dams were administered subcutaneous doses of dexmedetomidine 200 mcg/kg/day (equivalent to 1.8 times the intravenous MRHD of 17.8 mcg/kg/day based on body surface area [BSA]) during the period of organogenesis (Gestation Day [GD] 6 to 15). No malformations were reported.

No malformations or embryo-fetal toxicity were noted in a rabbit embryo-fetal development study in which pregnant does were administered dexmedetomidine intravenously at doses of up to 96 mcg/kg/day (approximately half the human exposure at the MRHD based on AUC) during the period of organogenesis (GD 6 to 18).

Reduced pup and adult offspring birth weights, and grip strength were reported in a rat developmental toxicology study in which pregnant females were administered dexmedetomidine subcutaneously at doses of 8 mcg/kg/day (0.07 times the MRHD based on BSA) during late pregnancy through lactation and weaning (GD 16 to postnatal day [PND] 25). Decreased viability of second-generation offspring and an increase in early implantation loss along with delayed motor development occurred in the 32 mcg/kg/day group (equivalent to less

than the clinical dose based on BSA) when first generation offspring were allowed to mate. This study limited dosing to hard palate closure (GD 15 to 18) through weaning instead of dosing from implantation (GD 6 to 7) to weaning (PND 21).

In a study in the pregnant rat, placental transfer of dexmedetomidine was observed when radiolabelled dexmedetomidine was administered subcutaneously.

Lactation

Risk Summary

Available published literature reports the presence of dexmedetomidine in human milk following intravenous administration (*see Data*). There is no information regarding the effects of dexmedetomidine on the breastfed infant or the effects on milk production. Advise women to monitor the breastfed infant for irritability. The developmental and health benefits of breastfeeding should be considered along with the mother's clinical need for Dexmedetomidine and any potential adverse effects on the breastfed infant from Dexmedetomidine or from the underlying condition.

Data

In two published clinical studies, a total of 14 women were given intravenous dexmedetomidine 6 mcg/kg/hour for 10 minutes after delivery followed by continuous infusion of 0.2–0.7 mcg/kg/hour. Breast milk and maternal blood samples were collected at 0, 6, 12, and 24 hours after discontinuation of dexmedetomidine. Plasma and milk dexmedetomidine concentrations were detectable up to 6 hours in most subjects, up to 12 hours in one subject and undetectable in all at 24 hours. The milk-to-plasma ratio from single paired maternal milk and plasma concentrations at each time point ranged from 0.53 to 0.95. The relative infant dose was estimated to range from 0.02 to 0.098%.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Patients should be advised to refrain from driving or other hazardous tasks for a suitable period of time after receiving dexmedetomidine for procedural sedation.

UNDESIRABLE EFFECTS

The following clinically significant adverse reactions are described elsewhere in the labelling: Hypotension, bradycardia and sinus arrest (*see Warnings and Precautions*).

Transient hypertension [*see Warnings and Precautions* (5.3)].

Clinical Trials Experience

Because clinical trials are conducted under widely varying conditions, adverse reactions rates observed in the clinical trials of a drug cannot be directly compared to rates in clinical trials of another drug and may not reflect the rates observed in practice.

Most common treatment-emergent adverse reactions, occurring in greater than 2% of patients in both Intensive Care Unit and procedural sedation studies include hypotension, bradycardia and dry mouth.

Intensive Care Unit Sedation

Adverse reaction information is derived from the continuous infusion trials of Dexmedetomidine for sedation in the Intensive Care Unit setting in which 1007 adult patients received Dexmedetomidine. The mean total dose was 7.4 mcg/kg (range: 0.8 to 84.1), mean dose per hour was 0.5 mcg/kg/hr (range: 0.1 to 6.0) and the mean duration of infusion of 15.9 hours (range: 0.2 to 157.2). The population was between 17 to 88 years of age, 43% ≥65 years of age, 77% male and 93% Caucasian. Treatment-emergent adverse reactions occurring at an

incidence of >2% are provided in Table 2. The most frequent adverse reactions were hypotension, bradycardia and dry mouth. (see *Warnings and Precautions*).

Table 2. Adverse Reactions with an Incidence > 2%- Adult Intensive Care Unit Sedation Population <24 hours*

Adverse Event	All Dexmedetomidine (N = 1007) (%)	Randomized Dexmedetomidine (N = 798) (%)	Placebo (N = 400) (%)	Propofol (N = 188) (%)
Hypotension	25%	24%	12%	13%
Hypertension	12%	13%	19%	4%
Nausea	9%	9%	9%	11%
Bradycardia	5%	5%	3%	0
Atrial Fibrillation	4%	5%	3%	7%
Pyrexia	4%	4%	4%	4%
Dry Mouth	4%	3%	1%	1%
Vomiting	3%	3%	5%	3%
Hypovolemia	3%	3%	2%	5%
Atelectasis	3%	3%	3%	6%
Pleural Effusion	2%	2%	1%	6%
Agitation	2%	2%	3%	1%
Tachycardia	2%	2%	4%	1%
Anemia	2%	2%	2%	2%
Hyperthermia	2%	2%	3%	0
Chills	2%	2%	3%	2%
Hyperglycemia	2%	2%	2%	3%
Hypoxia	2%	2%	2%	3%
Post-procedural Hemorrhage	2%	2%	3%	4%
Pulmonary Edema	1%	1%	1%	3%
Hypocalcemia	1%	1%	0	2%
Acidosis	1%	1%	1%	2%
Urine Output Decreased	1%	1%	0	2%
Sinus Tachycardia	1%	1%	1%	2%
Ventricular Tachycardia	< 1%	1%	1%	5%
Wheezing	< 1%	1%	0	2%
Edema Peripheral	< 1%	0	1%	2%

* 26 subjects in all Dexmedetomidine group and 10 subjects in the randomized Dexmedetomidine group had exposure for greater than 24 hours.

Adverse reaction information was also derived from the placebo-controlled, continuous infusion trials of Dexmedetomidine for sedation in the surgical intensive care unit setting in which 387 adult patients received Dexmedetomidine for less than 24 hours. The most frequently observed treatment-emergent adverse events included hypotension, hypertension, nausea, bradycardia, fever, vomiting, hypoxia, tachycardia and anemia (see Table 3).

Table 3: Treatment-Emergent Adverse Events Occurring in >1% of All Dexmedetomidine-Treated Adult Patients in the Randomized Placebo-Controlled Continuous Infusion <24 Hours ICU Sedation Studies

Adverse Event	Randomized Dexmedetomidine (N = 387)	Placebo (N = 379)
Hypotension	28%	13%
Hypertension	16%	18%
Nausea	11%	9%
Bradycardia	7%	3%
Fever	5%	4%
Vomiting	4%	6%
Atrial Fibrillation	4%	3%
Hypoxia	4%	4%
Tachycardia	3%	5%
Haemorrhage	3%	4%
Anaemia	3%	2%
Dry Mouth	3%	1%
Rigors	2%	3%
Agitation	2%	3%
Hyperpyrexia	2%	3%
Pain	2%	2%
Acidosis	2%	2%
Pleural Effusion	2%	1%
Oliguria	2%	<1%
Thirst	2%	<1%

In a controlled clinical trial, Dexmedetomidine was compared to midazolam for ICU sedation exceeding 24 hours duration in adult patients. Key treatment emergent adverse events occurring in dexmedetomidine, or midazolam treated patients in the randomized active comparator continuous infusion long-term intensive care unit sedation study are provided in Table 4. The number (%) of subjects who had a dose-related increase in treatment-emergent adverse events by maintenance adjusted dose rate range in the Dexmedetomidine group is provided in Table 5.

Table 4: Key Treatment-Emergent Adverse Events Occurring in Dexmedetomidine- or Midazolam-Treated Adult Patients in the Randomized Active Comparator Continuous Infusion Long-Term Intensive Care Unit Sedation Study

Adverse Event	Randomized Dexmedetomidine (N = 244)	Placebo (N = 122)
Hypotension ¹	56%	56%
Hypotension Requiring Intervention	28%	27%
Bradycardia ²	42%	19%
Bradycardia Requiring Intervention	5%	1%
Systolic Hypertension ³	28%	42%
Tachycardia ⁴	25%	44%
Tachycardia Requiring Intervention	10%	10%
Diastolic Hypertension ³	12%	15%
Hypertension ³	11%	15%

Hypertension Requiring Intervention [†]	19%	30%
Hypokalemia	9%	13%
Pyrexia	7%	2%
Agitation	7%	6%
Hyperglycemia	7%	2%
Constipation	6%	6%
Hypoglycemia	5%	6%
Respiratory Failure	5%	3%
Renal Failure Acute	2%	1%
Acute Respiratory Distress Syndrome	2%	1%
Generalized Edema	2%	6%
Hypomagnesemia	1%	7%

[†] Includes any type of hypertension.

¹ Hypotension was defined in absolute terms as Systolic blood pressure of <80 mmHg or Diastolic blood pressure of <50 mmHg or in relative terms as ≤30% lower than pre-study drug infusion value.

² Bradycardia was defined in absolute terms as <40 bpm or in relative terms as ≤30% lower than pre-study drug infusion value.

³ Hypertension was defined in absolute terms as Systolic blood pressure >180 mmHg or Diastolic blood pressure of >100 mmHg or in relative terms as ≥30% higher than pre-study drug infusion value.

⁴ Tachycardia was defined in absolute terms as >120 bpm or in relative terms as ≥30% greater than pre-study drug infusion value.

The following adverse events occurred between 2 and 5% for Dexmedetomidine and Midazolam, respectively: renal failure acute (2.5%, 0.8%), acute respiratory distress syndrome (2.5%, 0.8%), and respiratory failure (4.5%, 3.3%).

Table 5. Number (%) of Adult Subjects Who Had a Dose-Related Increase in Treatment Emergent Adverse Events by Maintenance Adjusted Dose Rate Range in the Dexmedetomidine Group

Dexmedetomidine mcg/kg/hr			
Adverse Event	≤0.7* (N = 95)	>0.7 to ≤1.1* (N = 78)	>1.1* (N = 71)
Constipation	6%	5%	14%
Agitation	5%	8%	14%
Anxiety	5%	5%	9%
Edema Peripheral	3%	8%	7%
Atrial Fibrillation	2%	4%	9%
Respiratory Failure	2%	6%	10%
Acute Respiratory Distress Syndrome	1%	3%	9%

* Average maintenance dose over the entire study drug administration

Procedural Sedation

Adverse reaction information is derived from the two trials for procedural sedation (see *Clinical Studies*) in which 318 adult patients received DEXMEDETOMIDINE. The mean total dose was 1.6 mcg/kg (range: 0.5 to 6.7), mean dose per hour was 1.3 mcg/kg/hr (range: 0.3 to 6.1) and the mean duration of infusion of 1.5 hours (range: 0.1 to 6.2). The population was between 18 to 93 years of age, ASA I-IV, 30% ≥65 years of age, 52% male and 61% Caucasian.

Treatment-emergent adverse reactions occurring at an incidence of >2% are provided in Table 6. The most frequent adverse reactions were hypotension, bradycardia, and dry mouth (*see Warnings and Precautions*). Pre-specified criteria for the vital signs to be reported as adverse reactions are footnoted below the table. The decrease in respiratory rate and hypoxia was similar between DEXMEDETOMIDINE and comparator groups in both studies.

Table 6. Adverse Reactions With an Incidence > 2%— Procedural Sedation Population

Adverse Event	Dexmedetomidine N = 318	Placebo N = 113
	n (%)	n (%)
Hypotension ¹	54%	30%
Respiratory Depression ²	37%	32%
Bradycardia ³	14%	4%
Hypertension ⁴	13%	24%
Tachycardia ⁵	5%	17%
Nausea	3%	2%
Dry Mouth	3%	1%
Hypoxia ⁶	2%	3%
Bradypnea	2%	4%

¹ Hypotension was defined in absolute and relative terms as Systolic blood pressure of <80 mmHg or ≤30% lower than pre-study drug infusion value, or diastolic blood pressure of <50 mmHg

² Respiratory depression was defined in absolute and relative terms as respiratory rate (RR)<8 beats per minute or >25% decrease from baseline

³ Bradycardia was defined in absolute and relative terms as <40 beats per minute or ≤30% lower than pre-study drug infusion value.

⁴ Hypertension was defined in absolute and relative terms as Systolic blood pressure >180 mmHg or ≥30% higher than pre-study drug infusion value or diastolic blood pressure of >100 mmHg.

⁵ Tachycardia was defined in absolute and relative terms as >120 beats per minute or ≥30% greater than pre-study drug infusion value.

⁶ Hypoxia was defined in absolute and relative terms as SpO2 < 90% or 10% decrease from baseline

Postmarketing Experience

The following adverse reactions have been identified during post approval use of Dexmedetomidine. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure. Hypotension and bradycardia were the most common adverse reactions associated with the use of Dexmedetomidine during post approval use of the drug.

Table 7: Adverse Reactions Experienced During Post Approval Use of Dexmedetomidine	
System Organ Class	Preferred Term
Blood and Lymphatic System Disorders	Anemia
Cardiac Disorders	Arrhythmia, ventricular arrhythmia, bradycardia, atrioventricular block, cardiac arrest, cardiac disorder, myocardial infarction, extrasystoles, atrial fibrillation, tachycardia, supraventricular tachycardia, ventricular tachycardia

Eye Disorders	Photopsia, visual impairment
Gastrointestinal Disorders	Abdominal pain, diarrhea, nausea, vomiting
General Disorders and Administration Site Conditions	Chills, hyperpyrexia, pain, pyrexia, thirst
Hepatobiliary Disorders	Hepatic function abnormal, hyperbilirubinemia
Investigations	Increased gamma-glutamyl transferase, alanine aminotransferase increased, aspartate aminotransferase increased, increased blood alkaline phosphatase, blood urea increased, electrocardiogram T wave inversion, electrocardiogram QT prolonged
Metabolic and Nutrition Disorders	Acidosis, hyperkalemia, hypoglycemia, hypovolemia, hypernatremia
Nervous System Disorders	Convulsion, dizziness, headache, neuralgia, neuritis, speech disorder
Psychiatric Disorders	Agitation, confusional state, delirium, hallucination, illusion
Renal and Urinary Disorders	Oliguria, polyuria
Respiratory Thoracic and Mediastinal Disorders	Apnea, bronchospasm, dyspnea, hypercapnia, hypoventilation, hypoxia, pulmonary congestion, respiratory acidosis
Skin and Subcutaneous Tissue Disorders	Hyperhidrosis, pruritus, rash, urticaria
Surgical and Medical Procedures	Light anesthesia
Vascular Disorders	Hemorrhage, blood pressure fluctuation, hypertension, hypotension

OVERDOSE AND TREATMENT

The tolerability of Dexmedetomidine was studied in one study in which healthy adult subjects were administered doses at and above the recommended dose of 0.2 to 0.7 mcg/kg/hr. The maximum blood concentration achieved in this study was approximately 13 times the upper boundary of the therapeutic range. The most notable effects observed in two subjects who achieved the highest doses were first degree atrioventricular block and second-degree heart block. No hemodynamic compromise was noted with the atrioventricular block and the heart block resolved spontaneously within one minute.

Five adult patients received an overdose of Dexmedetomidine in the intensive care unit sedation studies. Two of these patients had no symptoms reported; one patient received a 2 mcg/kg loading dose over 10 minutes (twice the recommended loading dose) and one patient received a maintenance infusion of 0.8 mcg/kg/hr. Two other patients who received a 2 mcg/kg loading dose over 10 minutes, experienced bradycardia and/or hypotension. One patient who received a loading bolus dose of undiluted Dexmedetomidine (19.4 mcg/kg), had cardiac arrest from which he was successfully resuscitated.

STORAGE CONDITION

Store below 30°C. Keep out of reach of children.

SHELF-LIFE

Shelf life before opening:

24 months

Shelf life after first opening:

The medicinal product must be used immediately after first opening.

Shelf life after dilution:

Chemical and physical in-use stability has been demonstrated for 24 hours at 25°C. From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to the use are the responsibility of the user and would not normally be longer than 24 hours at 2° C to 8°C, unless dilution has taken place in controlled and validated aseptic conditions.

LIST OF EXCIPIENTS

Sodium chloride USP q.s

Water for Injection USP q.s

DOSAGE FORMS AND PACKAGING AVAILABLE

The product is a 2 mL solution in clear 2R (3.4 mL) glass vial Type I, closed with 13mm, grey coloured, elastomeric fluoropolymer coated bromobutyl stopper Type I and 13mm green coloured, flip off cap made of aluminium with a plastic lid. 10 vials are then packed into a pre-printed outer carton along with a package insert.

PACK SIZE(S)

10 x 2 ml vials

MANUFACTURER

Fresenius Kabi Austria GmbH

Hafnerstrasse 36, Graz

8055, Austria

PRODUCT REGISTRATION HOLDER

Fresenius Kabi Malaysia Sdn Bhd

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DATE OF REVISION

March 2025

